Supplemental Material: Following the Information Bound from a Search-Window Prior with Sequential Bayesian Experiment Design in Nitrogen-Vacancy Magnetometry

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Equations, sections and figures without the prefix S refer to the main text.

S1. ODMR PHYSICS AND LINE SHAPES

This section gives the physics behind the line model of Sec. II A of the main text: the optical readout of the N-V spin, the Rabi response that sets the CW and pulsed line shapes, the hyperfine structure and the second transition that Eq. (4) leaves out, and the derivation of the line shapes with the laser treated as two rates rather than a reset, the broadening by the spin bath, and the pulsed readout.

A. Optical readout

The negatively charged N-V center is described by five levels [1, 2] (Fig. S1): the spin-triplet ground state $^3A_2$, split into $|0\rangle_g$ and $|\pm1\rangle_g$ by the spin projection m_s on the N-V axis; the spin-triplet excited state ^3E, with $|0\rangle_e$ and $|\pm1\rangle_e$; and a metastable spin singlet $|s\rangle$ that stands for the singlet manifold. Green light drives the spin-conserving optical transitions $|m_s\rangle_g \to |m_s\rangle_e$, which decay back by emitting red photoluminescence (PL). From $|\pm1\rangle_e$, however, the intersystem crossing to the singlet is much faster than from $|0\rangle_e$, and the singlet decays preferentially to $|0\rangle_g$. Two consequences follow. Under continuous illumination the spin is pumped into $m_s = 0$; and a center in $m_s = \pm1$ spends part of its optical cycles in the dark singlet, so it emits less PL than one in $m_s = 0$. The spin state can therefore be prepared and read out optically.

The two transitions $f_\pm$ share the state $|0\rangle_g$ but lie $2\delta_Z$ apart. When this separation is large compared with the linewidth, a drive near f_- is far off resonance with f_+ and couples only $|0\rangle_g$ and $|-1\rangle_g$. The excited state and the singlet then enter only through the rates at which the laser repolarizes the spin and destroys its coherence [3], so the spin reduces to a driven two-level system $\{m_s = 0, m_s = -1\}$.

B. Rabi response and the two ODMR modes

Let the drive have detuning $\Delta = f - f_-$ from the transition and Rabi frequency Ω, the rate in Hz at which the population oscillates on resonance. Off resonance the spin oscillates faster but less deeply, at the generalized Rabi frequency

$$\Omega_{\text{gen}} = \sqrt{\Omega^2 + \Delta^2}.$$

Starting from $m_s = 0$, the probability of finding the spin in $m_s = -1$ after a drive of duration t is

$$P_{-1}(\Delta, t) = \frac{\Omega^2}{\Omega_{\text{gen}}^2} \sin^2\!\big(\pi \Omega_{\text{gen}} t\big), \tag{S1}$$

and $P_0 = 1 - P_{-1}$ is the probability of finding it still in $m_s = 0$. The prefactor is the depth of the oscillation, the sine its time dependence. Since $m_s = -1$ is the darker state, the PL dip is proportional to P_{-1}. How the drive is applied sets the line shape.

CW ODMR. In continuous-wave ODMR the laser and the microwave drive act on the spin continuously and at the same time, and the PL is read out while the drive is on (Fig. S2). The laser keeps pumping the spin back into $m_s = 0$ while the drive keeps rotating it away, so every optical reset starts a new Rabi oscillation, and the PL, collected over many such cycles, reports Eq. (S1) averaged over the time T between resets. When the drive acts for many Rabi periods between resets, $\Omega_{\text{gen}} T \gg 1$, the oscillating factor averages to $\frac{1}{2}$,

$$\bar{P}_{-1}(\Delta) = \frac{\Omega^2}{\Omega_{\text{gen}}^2} \frac{1}{T} \int_0^T \sin^2\!\big(\pi \Omega_{\text{gen}} t\big)\, dt \;\longrightarrow\; \frac{1}{2} \frac{\Omega^2}{\Omega^2 + \Delta^2}. \tag{S2}$$

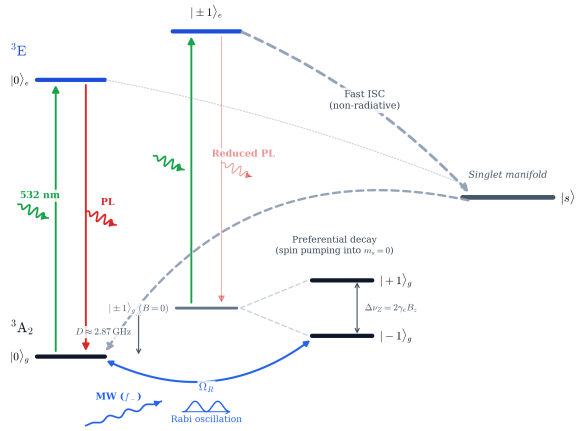


FIG. S1. Five-level model of the N-V center [1, 2]. [Caption to follow the figure.]

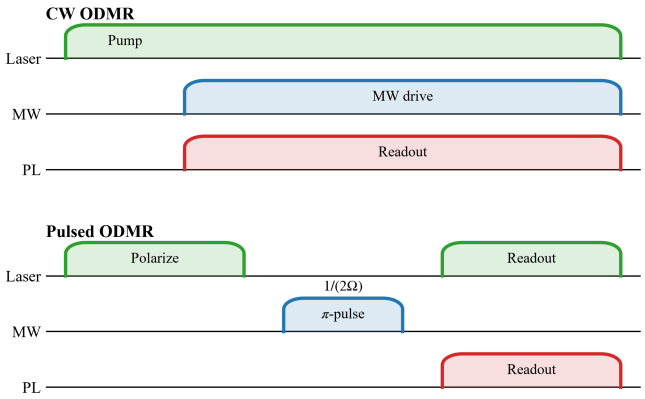


FIG. S2. Laser and microwave sequences of CW and pulsed ODMR. [Caption to follow the figure.]

The average only renormalizes the amplitude; the line shape is that of the prefactor, a Lorentzian in the detuning Δ of half width Ω. This is the microwave contribution to the width, power broadening: the stronger the drive, the further off resonance it still flips the spin. With the laser treated as two rates rather than a reset (Sec. S1 D), the line stays a Lorentzian, with the half width of Eq. (3) of the main text.

Pulsed ODMR is a technique that removes the laser from the width of the line [3]. In CW ODMR the laser that reads the spin out also broadens the line through Γ_c, and narrowing it means lowering both powers and with them the contrast and the PL. Pulsed ODMR separates the two in time: a laser pulse polarizes the spin into $m_s = 0$, the laser is switched off while a single microwave π-pulse of duration $t_\pi = 1/(2\Omega)$ is applied, and a second laser pulse reads the spin out (Fig. S2). The pulse fully inverts the spin on resonance, and Eq. (S1) at $t = t_\pi$ gives (Sec. S1 E)

$$P^{\pi}_{-1}(\Delta) = \frac{\Omega^2}{\Omega^2_{\text{gen}}} \sin^2\!\left(\frac{\pi}{2}\,\frac{\Omega_{\text{gen}}}{\Omega}\right). \tag{S3}$$

This line is not a Lorentzian. Its width is the Fourier limit of the pulse, $\sim 1/t_\pi$, down to $\sim 1/T_2^*$, where the spread of the transition frequency sets it, and it has side lobes up to 12% of the depth: dips where there is no resonance, a hazard for a search.

C. Terms left out of the line model

The Hamiltonian of Eq. (1) drops the strain term $E(S_x^2 - S_y^2)$, since it is a small correction when $\gamma_e B_z \gg E$.

Each nuclear projection m_I gives its own copy of the line, shifted by $A_\parallel m_I$ (Eq. (2)) and weighted by the population of that m_I, so the observed spectrum is a sum of $2I + 1$ lines. Two cases reduce it to the single line of Eq. (4). Near the excited-state level anticrossing the nitrogen nuclear spin is polarized into a single m_I [4]: only one line remains, and its fixed offset $A_\parallel m_I$ is absorbed into f_-. When the line is much wider than the hyperfine spacing, $\Gamma \gg |A_\parallel|$, the lines merge into one.

Equation (4) also leaves out the f_+ line. The drive window lies below D and holds f_- only, but the tail of the f_+ line, $2\delta_Z$ higher, adds $C\gamma^2/[\gamma^2 + (f - f_+)^2]$ inside it: at f_- a fraction $\gamma^2/(\gamma^2 + 4\delta_Z^2)$ of the contrast C, i.e. 20% at $\delta_Z = \gamma$ and below 1% for $\delta_Z \gtrsim 5\gamma$. Equation (4) assumes $\delta_Z \gg \gamma$; small Zeeman shifts, where the two lines overlap, are outside the scope of this work. The same tail raises the on-resonance depth to $C[1 + \gamma^2/(\gamma^2 + 4\delta_Z^2)]$, so the SNR C/σ of Eq. (8) is the $\delta_Z \gg \gamma$ limit of the on-resonance SNR.

D. CW line with optical pumping

Eq. (S2) treats the laser only as a reset, which is too crude when the resets are frequent. The laser acts on the spin through two rates, both set by how often the center goes through an optical cycle: it pumps the spin back toward $m_s = 0$, at a rate Γ_p, and each cycle destroys the phase between $m_s = 0$ and $m_s = -1$, at a rate Γ_c. The two rates share a cause but are not equal. A cycle that excites the center destroys the phase whatever happens next, since in the excited state the spin evolves under a different Hamiltonian and the decay path records its state; only the cycles that pass through the singlet move population into $m_s = 0$. Every pumping cycle therefore destroys the phase, but not every phase-destroying cycle pumps, and Γ_c exceeds Γ_p. The spin's environment adds rates of its own, independent of the laser: its longitudinal relaxation $1/T_1$ adds to Γ_p and its homogeneous dephasing $1/T_2$ adds to Γ_c, giving the total rates $\Gamma_1 = \Gamma_p + 1/T_1$ for the populations and $\Gamma_2 = \Gamma_c + 1/T_2$ for the coherence of Eq. (3). At the laser powers of CW ODMR both environmental rates are slow against the optical ones, $\Gamma_1 \approx \Gamma_p$ and $\Gamma_2 \approx \Gamma_c$, but the derivation below holds for any values of the totals. The slow spread of the transition frequency, which sets T_2^*, is not a rate of this kind and is treated separately below. The microwave enters only through its angular Rabi frequency $\Omega_R = 2\pi\Omega$. The competition between the two is clearest in terms of populations. A drive whose coherence is destroyed at the rate Γ_2 no longer produces Rabi oscillations; it moves population between $m_s = 0$ and $m_s = -1$, in both directions alike, at the rate

$$W(\Delta) = \frac{\Omega_R^2\,\Gamma_2}{2\left[\Gamma_2^2 + (2\pi\Delta)^2\right]} = \frac{\Omega_R^2}{2\Gamma_2}\,\frac{\Gamma_2^2}{\Gamma_2^2 + (2\pi\Delta)^2}, \tag{S4}$$

a Lorentzian of peak $\Omega_R^2/2\Gamma_2$ and half width $\Gamma_2/2\pi$ in Hz. The laser, helped by T_1, returns population from $m_s = -1$ at the rate Γ_1, so with $P_0 = 1 - P_{-1}$ the population of $m_s = -1$ obeys

$$\dot{P}_{-1} = W\,(P_0 - P_{-1}) - \Gamma_1 P_{-1} = W - (2W + \Gamma_1)\,P_{-1}.$$

In steady state, inserting Eq. (S4) and then ordering the terms of the denominator,

$$P_{-1}^{\rm ss}(\Delta) = \frac{W}{2W+\Gamma_1} = \frac{1}{2}\,\frac{\Omega_R^2\,\Gamma_2}{\Omega_R^2\,\Gamma_2 + \Gamma_1\big[\Gamma_2^2 + (2\pi\Delta)^2\big]} = \frac{1}{2}\,\frac{\Omega_R^2\,\Gamma_2/\Gamma_1}{(2\pi\Delta)^2 + \Gamma_2^2 + \Omega_R^2\,\Gamma_2/\Gamma_1}, \tag{S5}$$

the same as the steady state of the full Bloch equations [3]. It reduces to Eq. (S2) when the two rates are equal and small against Ω_R. The last form is a Lorentzian in $2\pi\Delta$ whose squared half width is the constant in its denominator, which gives Eq. (3).

The first term of Eq. (3) is the width without drive, set by the loss of coherence. The second is power broadening by the microwave: a strong drive saturates the transfer, $W \gg \Gamma_1$, already on the tails of the line, and the line widens until W there has fallen to about the pumping rate. The depth of the line, Eq. (S5) at $\Delta = 0$, is

$$P_{-1}^{\rm ss}(0) = \frac{1}{2}\,\frac{\Omega_R^2}{\Omega_R^2 + \Gamma_1\Gamma_2}\,:$$

it grows with the microwave power, toward $\frac{1}{2}$, and falls with the laser power, which repolarizes the spin before the drive can flip it.

Optical saturation. The two laser rates grow with the laser power, but not without limit. The optical transition is itself a two-level system: the laser excites the center at a rate proportional to its intensity $I_{\rm opt}$, and the excited state decays in its lifetime τ_e. In steady state the excited population is $s/(1+s)$, with the saturation parameter $s = I_{\rm opt}/I_{\rm sat}$ the ratio of the excitation rate to the decay rate $1/\tau_e$. At low power it grows linearly with s; at high power it levels off, because a center cannot be excited again before it has decayed. The rate of optical cycles is this population divided by τ_e, and both laser rates follow it, $\Gamma_{p,c} = \Gamma_{p,c}^{\infty}\, s/(1+s)$ [3], with $\Gamma_{p,c}^{\infty}$ their values at full saturation. Raising the laser power therefore both widens and flattens the line, up to saturation.

Spin bath in the CW line. The local fields of the spin bath, mostly the ^{13}C nuclear spins, shift the transition by a random offset ξ that changes slowly, over times much longer than the repolarization time $1/\Gamma_p$. The spin therefore reaches the steady state of Eq. (S5) at the detuning $\Delta - \xi$ before ξ changes, and a measurement that averages over many values of ξ records that steady state convolved with their distribution, the same average as Eq. (S6) below. For a Gaussian distribution of standard deviation $\sigma_\xi = 1/(\sqrt{2}\,\pi T_2^*)$ the CW line is a Voigt profile: the Lorentzian of Eq. (S5), of half width γ, convolved with a Gaussian of width σ_ξ. When $\gamma \gg \sigma_\xi$ it is the Lorentzian of Eq. (4); when the laser and microwave powers are lowered until $\gamma \lesssim \sigma_\xi$, the line stops narrowing at a width $\sim 1/T_2^*$, as the pulsed line does. In either case it is one dip with one total width, which is the width inferred in Sec. III of the main text; the simulations use a pseudo-Voigt approximation to this profile.

E. Pulsed line

The π-pulse length is chosen so that on resonance, where the spin oscillates at Ω, it is fully inverted: $\pi\Omega t_\pi = \pi/2$. Off resonance it oscillates at $\Omega_{\rm gen}$ instead, so the argument of the sine in Eq. (S1) at $t = t_\pi$ is $\pi\Omega_{\rm gen} t_\pi = (\pi/2)\,\Omega_{\rm gen}/\Omega$, which gives Eq. (S3). On resonance $\Omega_{\rm gen} = \Omega$ and the inversion is complete. For $|\Delta| \gg \Omega$, $\Omega_{\rm gen} \approx |\Delta|$ and $P_{-1}^{\pi} \approx \sin^2(\pi\Delta t_\pi)/(2\Delta t_\pi)^2$, the power spectrum of a rectangular pulse of length t_π: far from resonance the spin responds to the frequency content of the pulse. The line vanishes wherever $\Omega_{\rm gen} = 2\Omega, 4\Omega, \ldots$, i.e. at $|\Delta| = \sqrt{3}\,\Omega, \sqrt{15}\,\Omega, \ldots$: there the spin completes a whole number of full Rabi cycles during the pulse and returns to $m_s = 0$. Between the zeros lie side lobes that decay as $\Omega^2/\Omega_{\rm gen}^2$; the first, near $\Omega_{\rm gen} = 3\Omega$ at $|\Delta| \approx 2.7\,\Omega$, is 12% of the depth.

Inhomogeneous broadening. The transition frequency of the spin is not sharp: slowly fluctuating local fields, mostly from the bath of ^{13}C nuclear spins, shift it by a random offset ξ that is fixed during one pulse but changes from one repetition of the sequence to the next. A measurement averages many repetitions, each driven at the detuning $\Delta - \xi$ instead of Δ, so the measured line is Eq. (S3) averaged over the distribution of ξ, i.e. convolved with it. For a Gaussian distribution of standard deviation σ_ξ,

$$\bar{P}_{-1}^{\pi}(\Delta) = \int P_{-1}^{\pi}(\Delta - \xi)\,\frac{e^{-\xi^2/2\sigma_\xi^2}}{\sqrt{2\pi}\,\sigma_\xi}\,d\xi. \tag{S6}$$

The width σ_ξ is what T_2^* measures. A free spin precessing at the offset ξ picks up the phase $2\pi\xi t$, and the average of $e^{2\pi i\xi t}$ over the Gaussian is the coherence envelope $e^{-2\pi^2\sigma_\xi^2 t^2}$. T_2^* is defined by writing this envelope as $e^{-(t/T_2^*)^2}$, so $\sigma_\xi = 1/(\sqrt{2}\,\pi T_2^*)$. For $t_\pi \ll T_2^*$ the spread is small against Ω and the line is that of Eq. (S3). For $t_\pi \gtrsim T_2^*$ the Gaussian dominates: the zeros and side lobes are washed out, the line stops narrowing at a width $\sim 1/T_2^*$, and the depth falls because the pulse no longer inverts every frequency class [3].

Readout. While the microwave pulse is on, the laser is off and the spin relaxes only through T_1, which is much longer than t_π, and dephases through the spread already included in Eq. (S6). The readout is different: the laser that reads the spin out also repolarizes it, so the PL difference between $m_s = 0$ and $m_s = -1$ lasts only for the repolarization time τ_p. Taking the repolarization as a single exponential, a spin flipped with probability P_{-1} emits, a time t into the readout pulse, PL at the rate $\Phi_0\,[1 - c\,P_{-1}\,e^{-t/\tau_p}]$, where Φ_0 is the rate of $m_s = 0$ and c the fractional PL drop of $m_s = -1$ at the start of the readout. The recorded PL is this rate integrated over a readout window of length t_r, normalized to the same

integral with no flip,

$$\begin{aligned}\mu &= 1 - \frac{1}{t_r}\int_0^{t_r} c\,P_{-1}\,e^{-t/\tau_p}\,dt \\ &= 1 - c\,\frac{\tau_p}{t_r}\big(1 - e^{-t_r/\tau_p}\big)\,\bar{P}^{\pi}_{-1}(\Delta),\end{aligned} \tag{S7}$$

with $P_{-1} = \bar{P}^{\pi}_{-1}$ of Eq. (S6). The integration leaves the line shape unchanged and sets only its depth: the contrast is $C = c\,(\tau_p/t_r)(1 - e^{-t_r/\tau_p})$. A longer window collects more photons, $N_{\rm ph} \propto t_r$, but dilutes the contrast, so the single-shot signal-to-noise ratio $C\sqrt{N_{\rm ph}} \propto (1 - e^{-t_r/\tau_p})/\sqrt{t_r/\tau_p}$ is largest at $t_r \approx 1.26\,\tau_p$.

S2. THE WINDOW IN THE VAN TREES BOUND

For a prior density $p(\theta)$, the van Trees inequality [5, 6] bounds the mean squared error of any estimator, averaged over the prior, by

$$\sigma_\theta^2(n) \;\geq\; \big(J(p) + nI_{\max}\big)^{-1}, \qquad J(p) = \int \frac{p'(\theta)^2}{p(\theta)}\,d\theta.$$

For a Gaussian prior $J(p) = \sigma_0^{-2}$, the value used for tracking. A uniform window drops from $1/W$ to zero at its edges, so $J(p)$ is infinite and the inequality reduces to $\sigma_\theta \geq 0$: true, but empty. The inverse variance $12/W^2$ cannot simply take its place, since $J(p) \geq 1/\mathrm{Var}(\theta)$ with equality only for a Gaussian: for a signal linear in θ, $(12/W^2 + nI_{\max})^{-1/2}$ is the error of the best linear estimator, which an estimator can reach and the best estimator beats, so it is not a lower bound.

Worst-placed resonance. Instead of the prior, weight the positions in the window by $\rho(v) = (2/W)\cos^2(\pi v/W)$, with v the distance from the center. Of all densities confined to the window this one has the least Fisher information, $J(\rho) = 4\pi^2/W^2$ [6]. The van Trees inequality then bounds the mean squared error averaged with the weight ρ, and the worst-placed resonance is at least as bad as any such average, which gives Eq. (13) for a global search. The prior itself stays uniform; ρ is only a step in the argument. For a signal linear in θ the worst-placed resonance starts at $W/2$, π times above the bound, and the bound is tight at large n: with $m = \sqrt{nI_{\max}}\,W/2$ the half width in units of the measurement error, the best worst-case mean squared error is $(nI_{\max})^{-1}(1 - \pi^2/m^2)$ to order m^{-2}, with this same $\cos^2$ density as the least favorable one [7], and the bound agrees with this to that order.

Average over the window. This argument does not cover the error averaged over a uniformly placed resonance, which is what the runs report. For a signal linear in θ the best achievable average error can be computed numerically. It lies above the bound until $nI_{\max}W^2 \approx 400$, by a factor that starts at $\pi/\sqrt{3} \approx 1.8$ at $n = 0$ and falls below 1.1 by $nI_{\max}W^2 \approx 80$, that is $n \approx 2\,n_c$. From $nI_{\max}W^2 \approx 400$, $n \approx 10\,n_c$, it lies below the bound, by at most about 1% in σ, an effect that vanishes as n grows. The cause is the hard edge: the posterior of a resonance close to an edge is cut off there, so it is slightly easier to locate.

S3. EXPECTED INFORMATION GAIN ACROSS THE REGIMES

Sec. II F of the main text asks two questions of the variance form of the expected information gain, Eq. (18): when it agrees with the exact gain, and when maximizing it greedily puts the measurements where the van Trees bound needs them. This section gives the run-level argument that greedy choice is optimal in refine, and the argument for the two earlier regimes, where neither result holds.

When the variance form is exact. Eq. (18) is an equality when $\mu(d;\theta,\phi)$, with θ,ϕ drawn from the posterior, is Gaussian distributed. Whether it is depends on the posterior of θ, not on the noise: the noise is Gaussian by construction, while the posterior only becomes Gaussian with enough data (Sec. S4). Whether the Gaussian form is exact is therefore a question of whether the posterior makes the predicted signal Gaussian.

Greedy over the whole run. In any regime, the chain rule of mutual information [8] makes the gains of a run add up: on average over runs they equal what all the outcomes together taught about the parameters,

$$\mathbb{E}\Big[\sum_{i=0}^{n-1} \mathrm{EIG}_i(d_{i+1})\Big] = \underbrace{I(\theta,\phi;y_1,\ldots,y_n)}_{\text{learned from } n \text{ outcomes}}. \tag{S8}$$

This is bookkeeping and says nothing yet about the limit. The greedy rule, Eq. (19), maximizes each term given the ones before it. That also maximizes the sum only if where one measures now does not change what later measurements can gain. In refine this holds: each gain is the logarithm of one shrinkage, Eq. (20), so the gains from measurement m to n sum to $\ln(\sigma_{\theta,m}/\sigma_{\theta,n})$, and they depend on where the measurements were taken only through the sum of their $I(d_i)$. Maximizing every $I(d_i)$ therefore maximizes the total: in refine greedy EIG is optimal over the whole run and reaches the data limit of Eq. (13). The information the rule maximizes and the information the bound counts are then the same quantity.

Reading the posterior width. The conditions under which Eq. (20) holds, a single peak with $\sigma_\theta \ll \gamma$ and almost no mass elsewhere in the window, also decide when the posterior standard deviation can be read as a 68% interval: a posterior with a narrow peak and wide tails can have the same standard deviation as a Gaussian, with most of its mass much closer to the peak (Sec. S4).

Search: a two-valued signal. Before the dip is found the posterior of θ is spread over the window, and the predicted signal at d is two-valued: with probability p, the posterior mass that would put the dip at d, it is about

$1-C$, otherwise 1. Then $\sigma^2_{\rm pred} = p(1-p)C^2$, while the true mutual information is at most the binary entropy $H_b(p) \le \ln 2$. The Gaussian form is not exact: it keeps growing with SNR where the true gain saturates, so it overvalues search candidates relative to refine candidates; both peak at $p = \frac{1}{2}$, so they rank search candidates alike. Nor is greedy choice optimal over the run: the chain-rule argument of Eq. (S8) fails, since where one measures changes what later measurements can gain. Here greedy EIG is only myopically optimal, which is where the strategies of Sec. III B of the main text differ.

Localize: the variance form chases the tail. The same two-valued picture holds in the localize stage, with a different ranking. Once a peak has formed at the dip, a small residual mass p far from it gives a predictive variance $p(1-p)C^2$ at its location, large at high SNR even for small p, while the information it holds is at most $H_b(p)$, small for small p. The variance form therefore keeps spending measurements to remove such distant mass. This reduces the posterior variance, which the tail dominates (Sec. S4) and which is the quantity Eq. (13) constrains, rather than the posterior entropy: in the localize stage the variance rule acts as a design for the MSE, while a rule built on the information gain would refine the peak sooner and accept the tail. Whether this explains why localizing costs SBED far more measurements than the bound is a hypothesis we have not tested.

S4. APPROACH OF THE POSTERIOR TO A GAUSSIAN

Eq. (20) and the reading of σ_θ as a 68% interval assume a Gaussian posterior. This section states when that holds for the line of Eq. (4). There are two separate questions: whether the posterior has a single peak, and whether that peak is Gaussian. We take the nuisances as known; they add dimensions but do not change the scaling.

Shape of a single peak. Let $\ell(\theta) = \ln p(\theta) + \sum_i \ln \mathcal{L}_i(\theta)$ be the log posterior and $\hat\theta$ its maximum. Inside the window the uniform prior is flat and only the likelihood contributes. Expanding about the maximum (Laplace's method [9]),

$$\ell(\theta) = \ell(\hat\theta) - \frac{J}{2}(\theta-\hat\theta)^2 + \frac{\ell'''}{6}(\theta-\hat\theta)^3 + \frac{\ell''''}{24}(\theta-\hat\theta)^4 + \dots, \tag{S9}$$

with $J = -\ell''(\hat\theta)$, whose expectation is the Fisher information J_n of Sec. II B. Keeping only the quadratic term gives a Gaussian of variance $\sigma^2_\theta = 1/J$. Over the posterior, $|\theta - \hat\theta| \sim \sigma_\theta$, so the higher terms change the log density by relative amounts of order $\ell'''\sigma^3_\theta$ and $\ell''''\sigma^4_\theta$; the first sets the asymmetry of the peak, the second the weight of its tails.

For Gaussian noise, $\ln \mathcal{L}(\theta) = -k\,[y - \mu(d;\theta)]^2/2\sigma^2$ up to a constant. Averaged over the noise, the derivatives are

$$\mathbb{E}[\ell''] = -\sum_i \frac{k\,\mu_i'^2}{\sigma^2} = -J_n, \qquad \mathbb{E}[\ell'''] = -3\sum_i \frac{k\,\mu_i'\mu_i''}{\sigma^2},$$

where primes are derivatives with respect to θ at d_i. The line changes on the scale of its half width γ, so each further derivative brings a factor of order $1/\gamma$: $\mu'' \sim \mu'/\gamma$, hence $|\ell'''| \lesssim J_n/\gamma$ and $|\ell''''| \lesssim J_n/\gamma^2$. The corrections in Eq. (S9) are then of relative order

$$\ell'''\sigma^3_\theta \sim \frac{\sigma_\theta}{\gamma}, \qquad \ell''''\sigma^4_\theta \sim \frac{\sigma^2_\theta}{\gamma^2}. \tag{S10}$$

A single peak is Gaussian once it is narrow compared with the line, $\sigma_\theta \ll \gamma$. This is an upper estimate: at the steepest-slope points $\mu'' = 0$ (there $|\mu'|$ is at its maximum), so for measurements placed there the cubic term vanishes on average and the corrections start at order σ^2_θ/γ^2.

In the data limit, $J_n \approx n I_{\max}$ with $I_{\max}$ of Eq. (10), and with $\Gamma = 2\gamma$,

$$\frac{\sigma_\theta}{\gamma} \approx \frac{8}{3\sqrt{3}}\,\frac{1}{\mathrm{SNR}\sqrt{nk}} \approx \frac{1.5}{\mathrm{SNR}\sqrt{nk}},$$

so at an SNR of order 10 a few shots on the sides of the dip already make the peak Gaussian.

Linearization of the signal. The same ratio controls the step from a Gaussian posterior to a Gaussian prediction used in Eq. (20). Expanding the signal over the posterior, $\mu(d;\theta) = \mu(d;\hat\theta) + \mu'(d)(\theta-\hat\theta) + \frac{1}{2}\mu''(d)(\theta-\hat\theta)^2 + \dots$, the quadratic term relative to the linear one is $\mu''\sigma_\theta/2\mu' \sim \sigma_\theta/\gamma$. Eq. (20) therefore holds up to relative corrections of order σ_θ/γ, and of order σ^2_θ/γ^2 at the steepest slope.

Leftover mass. The expansion describes one peak. Early in the localize stage the posterior is instead a mixture: a peak of weight $1-p$ and width σ_θ at the dip, and a remainder of weight p spread over the part of the window not yet excluded, of width up to W. The variance of the mixture is

$$\mathrm{Var}(\theta) \approx (1-p)\,\sigma^2_\theta + p\,s^2_{\rm r},$$

where $s^2_{\rm r} \lesssim W^2/12$ is the mean squared distance of the remainder from the peak. The remainder dominates once $p \gtrsim \sigma^2_\theta/s^2_{\rm r}$, i.e. roughly $p \gtrsim 12(\sigma_\theta/W)^2$, a very small weight: for $W = 20\,\Gamma$ and $\sigma_\theta = 0.1\,\Gamma$ it is 3×10^{-4}. Until the remainder has fallen below this, the posterior standard deviation reports the remainder, not the error of the peak, and the posterior is far from Gaussian: a narrow peak with wide tails. Its entropy, by contrast, is changed by the remainder only by about $H_b(p) + p\ln(s_{\rm r}/\sigma_\theta)$, which is small for small p. This is the difference between the variance and the information gain discussed in Sec. II F.

The remainder is removed only by measuring where it would place the dip. A measurement within about γ of such a location, when the dip is not there, lowers

the log weight of that part of the remainder on average by the Kullback–Leibler divergence between the two predicted outcomes, $kC^2/2\sigma^2 = k\,\mathrm{SNR}^2/2$. Lowering a weight p to below $12(\sigma_\theta/W)^2$ takes, at each location, about $2\ln[pW^2/12\sigma_\theta^2]/(k\,\mathrm{SNR}^2)$ measurements, and at least one; a remainder spread over the whole window has of order W/Γ such locations. At high SNR the posterior therefore becomes Gaussian after about one pass over the remaining window, a cost that grows with W/Γ like that of the search. [Compare with when the SBED posterior becomes Gaussian in the runs (Results).]

S5. MEASUREMENTS TO A TARGET, BANDWIDTH AND SENSITIVITY OVER A RUN

This section derives the number of measurements a sweep and an adaptive strategy need to reach a target precision, from which the speedup of Eq. (22) and the bandwidth of Sec. II H of the main text follow, and the course of the sensitivity over a run.

Sweep. Where a sweep measures does not depend on what it has learned, so its expected information grows at the constant rate I_{sweep} of Eq. (21) from its first measurement, and the van Trees bound for the sweep is $\sigma_\theta \approx (\sigma_0^{-2} + nI_{\mathrm{sweep}})^{-1/2}$. This has the same $n^{-1/2}$ form as the refine regime of an adaptive strategy; the two differ in the rate, I_{sweep} against $I_{\max}$. A single sweep reaches this bound only once its posterior has one peak. Until the scan has passed over the dip it learns only that the dip is not where it has looked, so its error, like that of an adaptive run, stays near σ_0 and then drops when the dip is resolved, after about one pass of the window, or several at low SNR. What the sweep lacks is the phase after that: an adaptive strategy moves its measurements onto the slope, so its rate rises from about I_{sweep} to $I_{\max}$, while the sweep keeps spending a fraction of order Γ/W of its measurements on the line and stays at I_{sweep}. Averaged over many passes the steps of the individual runs are smoothed out, and the error follows the bound with no plateau and no cliff, as observed by Dushenko *et al.* [10].

Measurements to a target. This too is a model of the error, so the window enters through its inverse variance. An adaptive strategy pays the search cost n_d of the elimination model first and then gains $I_{\max}$ per measurement. Let σ_{tgt} (Hz) be the target precision, the standard deviation of f_- at which a run is considered done. The number of measurements to reach it is

$$N_{\mathrm{sweep}}(\sigma_{\mathrm{tgt}}) = \frac{\sigma_{\mathrm{tgt}}^{-2} - \sigma_0^{-2}}{I_{\mathrm{sweep}}},$$
$$N_{\mathrm{ad}}(\sigma_{\mathrm{tgt}}) = n_d + \frac{\sigma_{\mathrm{tgt}}^{-2} - 12/\Delta d^2}{I_{\max}}, \tag{S11}$$

where the adaptive count follows the elimination model, Eqs. (15)–(16): n_d measurements to detect the dip, after which the information starts at $12/\Delta d^2$ and grows by $I_{\max}$ per measurement until it reaches $\sigma_{\mathrm{tgt}}^{-2}$. The sweep count uses the averaged rate, so it holds for targets that take more than one pass of the window. The speedup is $R(\sigma_{\mathrm{tgt}}) = N_{\mathrm{sweep}}/N_{\mathrm{ad}}$. For a precise target the refine terms dominate and R tends to $R_\infty = I_{\max}/I_{\mathrm{sweep}}$ of Eq. (22). For a loose target the search cost n_d dominates N_{ad} and $R(\sigma_{\mathrm{tgt}}) < R_\infty$, so the full dependence on window, linewidth, SNR (through n_d at low SNR) and target is given by Eq. (S11).

Bandwidth. The time to reach the target is t_m times $N_{\mathrm{ad}}(\sigma_{\mathrm{tgt}})$, n_d measurements to find the dip and the rest to refine it:

$$T(\sigma_{\mathrm{tgt}}) = t_m\Big[n_d + \frac{\sigma_{\mathrm{tgt}}^{-2} - 12/\Delta d^2}{I_{\max}}\Big]. \tag{S12}$$

For a tight target the refine term dominates, $T \approx t_m\sigma_{\mathrm{tgt}}^{-2}/I_{\max}$, which is the sensitivity η of Eq. (24) again. For a loose target the search term dominates, $T \approx n_d t_m$: the bandwidth cannot exceed about $1/(n_d t_m)$, whatever the SNR, which is $2\Delta d/(Wt_m)$ for a resonance halfway through the window.

Sensitivity as a function of time. Eq. (24) is a limit. Read off a run of length T, the product $\eta(T) = \sigma_B(T)\sqrt{T}$ also carries the search, and its expected course follows from the error curve of Sec. II D with $n = T/t_m$. During the search the error barely falls, Eq. (15), so the product grows,

$$\eta(T) = \frac{1}{\gamma_e}\,\frac{W - (T/t_m)\Delta d}{\sqrt{12}}\,\sqrt{T} \;\propto\; \sqrt{T}, \qquad T < n_d t_m.$$

Detection drops it by the localize factor, and afterwards, by Eq. (16),

$$\frac{\eta(T)}{\eta} = \Big(1 - \frac{n_e\,t_m}{T}\Big)^{-1/2} \approx 1 + \frac{n_e\,t_m}{2T}, \qquad T \geq n_d t_m, \tag{S13}$$

with η the sensitivity of Eq. (24) and $n_e = n_d - 12/(\Delta d^2 I_{\max})$. The measurements spent on the search, less the information that detection supplies, are a fixed debt n_e; refine pays it off, so $\eta(T)$ approaches η from above and the excess falls as $1/T$ (Fig. S3). A run therefore reports a sensitivity within a fraction ϵ of η only once $T \gtrsim n_e t_m/2\epsilon$, i.e. after about $50\,n_e$ measurements for 1%. For any strategy described by (n_0, q), Eq. (23) gives the same form, $\eta(T) \approx (\eta/\sqrt{q})\,(1 - n_0 t_m/T)^{-1/2}$: q sets the floor and n_0 how slowly it is reached. The unbiased Cramér–Rao bound $(nI_{\max})^{-1/2}$ is the constant η at all T; the van Trees bound, which includes the prior, lies below it and rises to it as $\sqrt{T}$ while $T \lesssim n_c t_m$, but no strategy can follow it there, since J_0 belongs to the window and not to any measurement.

S6. ALGORITHM SETTINGS

Particle filter. The cloud has $N_p = 1000$ particles. Every dimension is scaled to $[0, 1]$ by its prior bounds, so

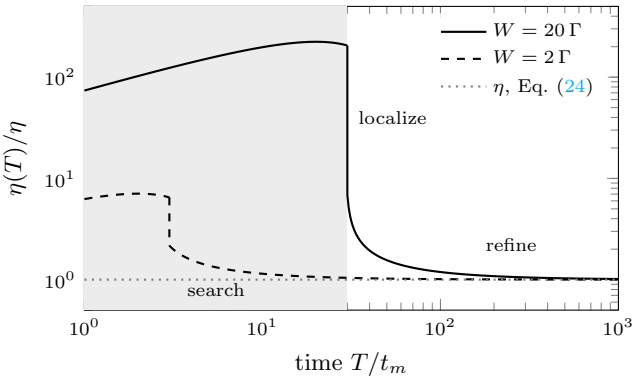


FIG. S3. Expected sensitivity $\eta(T) = \sigma_B(T)\sqrt{T}$ of an adaptive strategy as a function of the run time, from the elimination model with the values of Fig. 1. Solid: wide window, $W = 20\,\Gamma$; dashed: narrow window, $W = 2\,\Gamma$; dotted: the refine limit η of Eq. (24), which is also the unbiased Cramér–Rao bound. During the search (shaded) $\eta(T)$ grows as $\sqrt{T}$; detection drops it, and in refine it approaches η from above with an excess $n_e t_m/2T$, Eq. (S13). Simulated and measured runs are shown against η in Sec. IV.

one resampling kernel serves all of them; reported values are mapped back to Hz. The cloud is resampled when the effective sample size $(\sum_i w_i^2)^{-1}$ falls below $0.2\,N_p$, systematically, and then rejuvenated with the Liu–West kernel,

$$\psi_i' \sim \mathcal{N}\big(a\psi_i + (1-a)\bar{\psi},\ (1-a^2)\Sigma\big), \qquad a = 0.98, \quad \text{(S14)}$$

with $\psi_i = (\theta_i, \phi_i)$ the whole particle and $\bar{\psi}$, Σ the weighted mean and covariance before resampling. The diagonal of Σ is floored at $(0.01\,e^{-n/25})^2$ of each unit range after n measurements, and a particle that leaves the unit cube is reflected back into it.

Noise model. Each particle carries an inverse-gamma posterior on σ^2 with shape α, the same for every particle because its update does not depend on the residuals, and scale β_i. The prior is $\alpha_0 = 10$, $\beta_0 = \alpha_0\sigma_{\text{nom}}^2$, with σ_{nom} the geometric mean of the prior range of σ. Before each measurement both are discounted by $d = 0.99$; a measurement y_n, the mean of k shots at d_n, then adds $\frac{1}{2}$ to α and $\frac{1}{2}k\,r_{n,i}^2$ to β_i, with residual $r_{n,i} = y_n - \mu(d_n; \theta_i, \phi_i)$, so that after n measurements $\beta_i = d^n\beta_0 + \frac{1}{2}\sum_{t<n} d^{n-1-t} k\, r_{t,i}^2$. Old residuals are thus forgotten with a memory of about $1/(1-d) = 100$ measurements, and α saturates at $0.5/(1-d) = 50$, about 100 degrees of freedom. With σ^2 integrated out, the measurement reweights the particles by the Student-t predictive

$$w_i \ \propto\ w_i\, t_{2\alpha}\Big(y_n;\ \mu(d_n;\theta_i,\phi_i),\ \frac{\beta_i}{k\alpha}\Big), \qquad \text{(S15)}$$

taken with α, β_i before the update. After a rejuvenation each β_i is recomputed from the full history at the particle's new position.

Initial cloud. The simulated runs are global searches (Sec. II C): the prior on the Zeeman shift δ_Z is uniform over the range the truth is drawn from. [Tracking runs, with a Gaussian prior and the truth drawn from that same Gaussian, are not simulated yet.]

Stop rule and budget. The ceilings on the claimed uncertainty are 100 kHz for θ, 500 kHz for each width and 0.01 for the contrast; they must hold for 8 consecutive measurements, together with a dip found by the dip test, which is re-run at most every 10 measurements. Every strategy has the same budget of 2500 measurements; a sweep's total is capped in whole passes. A run that reaches the budget without stopping is censored and counted at the budget, as a lower bound on its cost.

Dip test. Each measurement gets the score $z_n = (b - y_n)/s$. The baseline b is the 70th percentile of the y_n minus its expected offset under pure noise, $t_\nu^{-1}(0.7)\,s$, and s and ν are the scale and degrees of freedom of the noise posterior. Every contiguous stretch of measured drive frequencies whose span lies in the prior range of the linewidth gets the boxcar score $\sum z_n/\sqrt{n}$, over its n measurements. A stretch is significant when its score exceeds $t_\nu^{-1}(1 - p_{\text{FA}}/N_{\text{tests}})$, the Bonferroni threshold for a family-wise false-alarm probability $p_{\text{FA}} = 0.1$ over the N_{tests} stretches tested.

Sweep family. Γ_0 is the FWHM of the prior-mean line. SimpleSweep measures a uniform grid of spacing Γ_0/p, $p = 2, 3, 4$. Smart-Sweep sweeps at $\Gamma_0/3$ until the dip test finds a dip, then at $\Gamma_0/10$ inside the focus, and then measures only at the maxima of $(\partial\mu/\partial\theta)^2$ at the posterior mean, cycling between them, once the posterior expects each of those points to carry at least 90% of the information of the true optimum.

Bayesian SBED. After every resampling SBED places 800 candidate drive frequencies at the quantiles of a density with Gaussian kernels on the two sides of the posterior-mean dip, ±HWHM from its center, kernels on the dips found and a flat share, thinned to a spacing of at least 100 kHz. Eq. (18) is evaluated on a 500-particle subsample, with σ^2 the posterior mean of the noise. The candidates are taken in blocks of 64, and the block maxima compete through a softmax with a temperature of 0.01 nats, so the choice is near-greedy. Each focus section is a dip's extent padded by one posterior-mean HWHM on each side. A new dip widens the focus at once; narrowing waits 8 measurements and must shrink the focus by at least 5%. The focus is re-decided after every resampling. The falsification probe is made with probability $0.02 + 0.08\,g$, where g is the widest unmeasured gap as a fraction of the window.

Use cases. The simulated signals are pseudo-Voigt dips of total FWHM 2 MHz, on a grid of the SNR, the window (1.6, 2, 3 and 5 times the width one dip group covers), the Gaussian share of the width (0, 0.15, 0.3) and the isotope (unresolved hyperfine lines, ^{14}N or ^{15}N). The code labels the SNR levels 1, 2, 5, 10, 20 against a contrast of 0.25 [convert to C/σ of Eq. (8): the code's contrast is $2C$]. [Number of repeats per use case.]

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